Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial – 6: Part (b)

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(1) (Change of Scale Property). If $\mathcal{L}\{f(t)\} = F(s)$, then show $\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$ for $a > 0$.

(2) Consider the following functions:

(i) The floor function $f(t) = \lfloor t \rfloor$, where for any integer n, $\lfloor t \rfloor = n \quad$ for all $n \leq t < n+1$.

(ii) $f(t) = \frac{e^{t^2}}{e^{2t}+1}$.

(iii) $f(t) = t^2 e^{-t}$.

Answer the following questions for each function.

(a) Is $f(t)$ continuous on $0 \leq t < \infty$, discontinuous but piecewise continuous on $0 \leq t < \infty$, or neither?

(b) Do there exist fixed numbers a and M such that $|f(t)| \leq Me^{at} \quad$ for $0 \leq t < \infty$?

(3) Find the Laplace transforms of the following functions

(a) $t^m \cosh(bt)$

(b) $e^t \sin(at)$

(c) $t\, e^t \sin(at)$

(d) $\dfrac{\sin t \cosh t}{t}$

(e) $5e^{-7t} + t + 2e^{2t}$

(f) $f(t) = \begin{cases} \sin 3t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$

(4) Find the inverse Laplace transforms of the following functions:

(a) $\tan^{-1}\left(\dfrac{a}{s}\right)$

(b) $\log\left(\dfrac{s^2+1}{(s+1)^2}\right)$

(a) $\frac{1-3s}{s^2+8s+21}$

(c) $\dfrac{s+2}{(s^2+4s-5)^2}$

(d) $\dfrac{se^{-\pi s}}{s^2+4}$

(e) $\dfrac{(1-e^{-2s})(1-3e^{-2s})}{s^2}$

(5) Use Laplace transforms to solve the following initial value problems.

(a) $y'' + 4y = \cos 2t, \qquad y(0) = 0, \quad y'(0) = 1.$

(a) $y'' + 4y' = \cos(t-3) + 4t, \qquad y(3) = 0, \quad y'(3) = 7.$

(b) $y'' + 3y' + 2y = f(t), \qquad y(0) = 0, \quad y'(0) = 0,$ where

$$f(t) = \begin{cases} 4t, & 0 < t < 1, \\ 8, & t > 1. \end{cases}$$

(c) $y'' + 4y = f(t), \qquad y(0) = 4, \quad y'(0) = 4,$ where

$$f(t) = \begin{cases} 8\sin t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$

(6) Find the Laplace transform of the sawtooth curve: